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Agentic AI and the rise of in silico team science in biomedical research

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Agentic artificial intelligence (AI) systems are emerging as teams of intelligent computational experts capable of rivaling human performance in labor-intensive tasks, including literature review, hypothesis formulation, data analysis and model interpretation. These systems are poised to accelerate labor-intensive biomedical research by making autonomous decisions based on contextual information and expert feedback. Agentic AI systems have been developed for various applications, including drug discovery, data analysis and biomarker identification; however, several distinct challenges remain for making these systems broadly deployable in biomedical research. Here we discuss three key algorithms and seven foundational building-block characteristics that contribute to the development of agentic AI systems. We highlight their biomedical applications, design considerations and the challenges and opportunities associated with deploying agentic AI systems to advance collaborative scientific research.

Recent advances in artificial intelligence (AI) technologies are reshaping traditionally labor-intensive tasks such as information retrieval (for example, ChatGPT), coding (for example, GitHub Copilot) and image generation (for example, DALL-E). These developments, driven by substantial improvements in algorithms, such as large language models (LLMs) and reinforcement learning (RL), have reignited interest in agents and agentic AI.

In this work, we define an agent as a system comprising one or more computational models that operate autonomously to achieve human-specified goals by processing data to guide solution generation, decision-making and execution. In general, computational systems leverage model outputs to determine appropriate courses of action and achieve user-specified goals. Traditional computational models, such as logistic regression, are often trained to select actions from a predefined, finite set, which limits their ability to extrapolate to new domains and actions. In parallel, traditional computational systems rely on rigid, user-specified conditions that enforce fixed workflows of these models. In contrast, agents can process and generate information flexibly, enabling them to generalize beyond predefined actions and execute complex operations. Ultimately, agents reduce the need for rigid, user-defined workflows by autonomously generating dynamic decision-making processes, allowing these systems to address complex and challenging problems. In the future, we expect these agents to learn and update their knowledge (for example, model weights) from past experiences and the current environment and to extrapolate to new domains and problems.

Agentic AI refers to a framework that enables collaboration among one or more autonomous agents to pursue a shared high-level objective. Agentic AI holds considerable promise for transforming activities long considered as uniquely human. One such activity is biomedical research, which typically involves iterative and repetitive processes such as literature review, hypothesis generation, experiment design, data analysis, result interpretation and real-world validation. Multiple factors impede biomedical research, including limited availability of experts, time constraints, limited funding resources and high demands for diverse skills and knowledge. Agentic AI systems are beginning to demonstrate potential as effective tools for mitigating some obstacles in biomedical research by creating domain-specific expert agents, processing a corpus of scientific literature and iteratively updating experimental plans (Fig. 1a).

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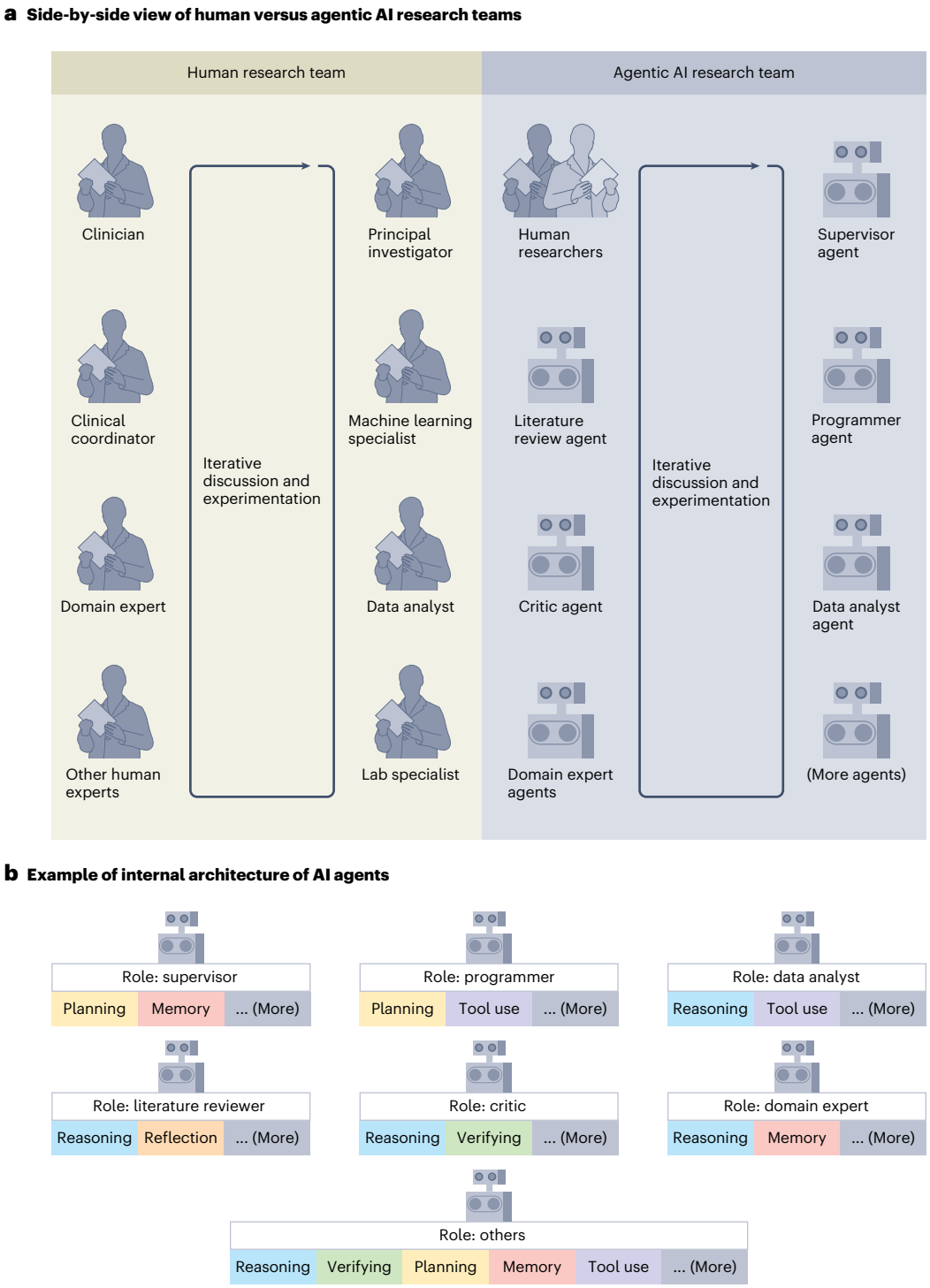


Fig. 1 | Human versus agentic AI research teams and the characteristics of role-playing AI agents in biomedical research. Biomedical research often demands collaboration among various domain experts from multiple human research teams. Additionally, it usually involves iterative experiments and analyses that require substantial domain knowledge. Agentic AI systems are beginning to show promise in addressing these skills and domain knowledge demands by creating agents with the domain expertise needed to complete complex tasks. **a**, Comparison between a human research team and an agentic AI one. **b**, Examples of the internal characteristics of role-playing AI agents. Each AI agent can be assigned specific roles and duties and enhanced with the necessary AI characteristics. With the rise of multi-agent AI systems, the quality of communication across human researchers, AI agents and computational tools will influence the achievement of the final research goal set by human researchers.

Although biomedical agentic AI remains at an early and formative stage—with many demonstrations reported as preprints or technical reports rather than fully peer-reviewed studies—ongoing advances in AI algorithms and engineering techniques are rapidly expanding its capabilities. In this study, we reflect on these developments and discuss how agentic AI systems can be responsibly and effectively deployed across diverse domains of biomedical research. We first discuss three relevant AI algorithms that drive agentic AI, followed by seven key characteristics that serve as the building blocks for agentic AI systems. Next, we highlight existing biomedical applications and benchmarks

for agentic AI systems and then discuss biomedical research-specific challenges, including data complexity, privacy, security, cost and fairness. Finally, we outline the opportunities that agentic AI systems offer for the future of biomedical research.

Key algorithms driving agentic AI

The concept of an 'agent'[1] has a long history in computer science, dating back to Alan Turing's 'thinking machines'[2] and has found wide applications across various biomedical fields, including epidemiology[3] and cancer biology[4]. Here we focus on LLM-based agents and on how agentic AI systems composed of such agents can facilitate biomedical discoveries through textual exchanges with human researchers.

In recent years, the rapid development of LLMs has accelerated the idea of AI assistants capable of following human instructions. Researchers have observed emergent behaviors in LLMs that resemble reasoning. For example, LLMs can perform multistep problem solving when prompted, despite not having been explicitly trained for task decomposition[5]. These observations have shifted perceptions of LLMs in both academia and industry—from static text-completion tools to dynamic decision-making agents—and have contributed to the emergence of LLM-based agentic AI systems. RL has subsequently attracted considerable attention for its ability to align LLM responses with human preferences[6] or reasoning strategies[7]. Large reasoning models (LRMs), which are LLMs further trained through RL for multistep reasoning, marked the fruition of this shift in perspective. Today, LLMs or LRMs (for example, OpenAI's GPT-5.2, Anthropic's Claude Opus 4.5, Google's Gemini 3, DeepSeek-V3.2 (ref. 8) and Mistral's Magistral[9]) serve as the primary verbal reasoning engines. Agentic AI systems are further defined by their system architectures, which include components such as tools and memory. In this context, evolutionary algorithms have been introduced as an approach to evolve and optimize responses or architectural designs for agentic AI systems, depending on the task at hand.

Agentic AI remains an active and rapidly evolving field. In this study, we focus on three algorithms—LLMs, RL and evolutionary algorithms—that contribute to the development of agentic AI systems.

Deep learning and LLMs

Deep learning uses multilayer neural networks to learn abstract representations of data for downstream prediction tasks[10]. LLMs are not agentic AI systems per se; rather, they are deep learning models trained on large-scale text corpora to perform next-token or masked-token prediction.

Models such as GPT use preceding context within a sentence to predict the next word, whereas models such as BERT[11] leverage complete sentence context to predict masked tokens. LLMs can also be categorized based on the public availability of their model weights, training data or both. Proprietary systems such as GPT-4 fall into the closed-source, closed-weight category. Models such as DeepSeek-V3.2 (ref. 8) and Llama 4 are open-weight but not fully open-source initiatives, as they release model weights while relying on proprietary training data, infrastructure and optimizations. A third category, often supported by government funding or fully open-source efforts, provides model weights, source code and training data under open licenses, enabling broad public research and use. Open-source and open-weight LLMs, exemplified by BLOOM[12], introduce additional considerations related to funding structures and digital sovereignty.

An LLM (for example, GPT-5) instance often serves as the primary driver of many agentic AI systems, translating human instructions into computational actions. LLMs are prone to hallucination, a phenomenon in which they generate nonsensical or false responses. Various techniques can reduce but not eliminate LLM hallucinations due to LLM's nature as probabilistic models[13,14]. However, because of their unique ability to express logic in natural languages, LLMs continue to serve an important role in agentic AI systems.

Reinforcement learning

RL is a widely used approach for training and improving agentic AI systems, with increasing applications in medical and biomedical research[15]. The key components of RL include the agent, environment, state, reward and policy[16]. Through continuous interactions with its environment, an RL agent learns to optimize cumulative rewards.

The reward-driven nature of RL allows researchers to design reward functions that incorporate human feedback or ethical principles during training[17]. For example, Biomni-R0, a biomedical agentic AI system, uses a composite reward that combines compliance with human-preferred response formats and accuracy against expert-annotated ground truth, thereby iteratively learning reward-maximizing actions.

The pivotal role of rewards in RL, however, poses unique challenges for evaluation. These include vulnerability to reward hacking—where an agent exploits ambiguities in the reward function to achieve high rewards without accomplishing the intended task—limited availability of expert-annotated outcome labels, inaccurate reasoning process despite correct outcomes and failure to reward correct intermediate steps[6,18–21]. To mitigate these issues, recent work has proposed approaches such as intermediate reward models to ensure correct reasoning trajectories. When intermediate processes can be verified through objective tests (for example, ground-truth validation), the approach is classified as RL with verifiable rewards (RLVR[22]). When rewards can only be assessed through preferences, it is classified as preference-based RL, including RL with human feedback (RLHF[17,21]) or with AI feedback[23]. These RL methods can be combined depending on the nature of the rewards and the availability of human experts for process-level or outcome-level supervision. Evaluating RL algorithms remains challenging in biomedical applications due to safety, ethics and cost concerns[15].

Evolutionary algorithms

Inspired by biological evolutionary principles, evolutionary algorithms are population-based optimization techniques that iteratively refine candidate solutions. Typically, an evolutionary algorithm begins by initializing a population of random candidate solutions, evaluating each on the target problem and selecting the highest-performing solutions as parents. These parents then undergo variation to generate the next generation of candidate solutions. The newly generated solutions undergo the same evolutionary cycle until a stopping criterion is met.

The integration of evolutionary algorithms within agentic AI is relatively recent but promising (for example, in enhancing code-generating agents[24–26]). As they frequently identify solutions that differ from human intuitions, they offer a powerful means of discovering new agent behaviors and architectures.

Agentic AI characteristics

We deem seven characteristics critical to the development of agentic AI systems tailored for biomedical research purposes (Table 1).

Reasoning

Reasoning in agentic AI refers to the process of deriving conclusions from ingrained knowledge and contextual information by emulating human cognitive strategies and applying formal logical reasoning techniques[27]. The ability of agentic AI systems to articulate their decisions (for example, recommending specific computational methods over others) is crucial for researchers to evaluate the appropriateness of agents' chosen courses of action and provide necessary interventions. As these reasoning capabilities advance, LLM-based agentic AI systems may function with reduced human supervision when addressing analytical tasks in biomedical research.

Fine-tuning pretrained LLMs improves their ability to answer complex scientific or mathematical questions that require reasoning[28–30]. However, fine-tuning is computationally demanding and may yield diminishing performance gains[31].

Table 1 | Comparison of agentic AI characteristics with the task of solving mathematical multiplication (2×3×4)

Characteristics	Description	Example
Reasoning	Generate conclusions from existing knowledge and/or input.	2×3×4 = (2×3)×4=6×4=24
Verification	Check whether the output and procedures are correct.	Are all intermediate steps satisfied, (2×3=6) and (4×6=24)?
Reflection	Identify and improve instruction/feedback/mistakes.	I got 2×3×4=26, but another expert agent got 24. What have I done wrong here?
Planning	Decompose tasks and organize actions.	Step 1, calculate 2×3=*X*; step 2, *X*×4=*Y*; step 3, use a calculator to verify the answer Y.
Tool use	Access and use the appropriate tool.	Search web documents. Write a Python code to generate the answer. Search in memory how 26 is wrongfully derived.
Memory	Memorize, summarize and/or forget past events.	Recall at which step the experiment veers off.
Communication	Effective conversations with humans or other agents.	JSON {'task type': 'arithmetic', 'operation': 'multiply', 'operands': [2, 3, 4]}

Inference-time techniques, such as in-context learning and prompt engineering, provide cost-effective alternatives to fine-tuning that enhance the reasoning capabilities of agentic AI systems without additional training. In-context learning elicits reasoning that follows a specific pattern by including examples within prompts provided to LLMs[32–34]. Prompt engineering designs prompts to guide LLM behaviors. Instead of producing a spontaneous response, chain-of-thought provokes multistep reasoning from LLMs, thereby improving performance on complex questions[5]. Other prompt engineering techniques, such as Tree-of-Thought[35] and Graph-of-Thoughts[36], extend beyond the linear reasoning paradigm of chain-of-thought[5] by allowing LLMs to produce a more complex interconnected network of thoughts.

Inference-time techniques, however, do not guarantee robust reasoning. Agentic AI systems may still experience reasoning cliffs and performance degradation when solving high-complexity problems[37]. Methods that improve robustness, such as adding pause tokens to induce iterative self-reasoning[38], must be incorporated during LLM pretraining or fine-tuning rather than applied ad hoc at inference. Consequently, the true effectiveness of inference-time techniques remains uncertain without complexity-controlled evaluation metrics for biomedical applications.

Agentic AI systems have not yet surpassed human researchers in domain-specific reasoning. For example, PubMedQA's best-performing model—a fine-tuned BioBERT LLM—achieves 68% accuracy, compared with 78% accuracy for human respondents on PubMed abstract–based questions[39]. Therefore, the ability of agentic AI systems to reason about domain-specific questions, particularly in interdisciplinary biomedical research, warrants further investigation.

Verifying

Verifying aims to ensure the correctness of both the reasoning process and the final answer by aligning them with factual knowledge. Unlike reasoning, verifying is not generative. Instead, it attempts to mitigate LLM hallucination by rewarding accurate intermediate steps and final outputs. Approaches, such as adding a result-verification layer for outputs in fine-tuning[40] or deploying a combination of outcome-level RLVR and process-level RLHF[20,21], supervise an agent's output and procedural correctness during training.

Verifying is a labor-intensive task that requires substantial high-quality human labeling. Math-Shepherd[41], a process-level RLVR method, demonstrates how intermediate steps in mathematical proofs can be automatically verified without human supervision. While formal mathematical proofs are not commonly encountered in biomedical research, the capacity to construct a valid proof provides a valuable proxy for assessing an agent's ability to reason systematically within a well-defined set of mathematical rules. An agent's success in biomedical research tasks may be evaluated based on the final and procedural correctness or alignment with objective scientific knowledge, similar to correctly deriving mathematical proofs.

Reflection

Reflection enhances reasoning through iterative self-improvement. There have been various inference-time reflection techniques for LLMs. Reflexion[42] self-improves by critiquing and memorizing each step it implements, allowing the system to analyze why an approach fails and to avoid repeating the same mistakes. CRITIC[43] resorts to external tools, such as knowledge bases and computational programs, to critique and improve a response. Self-Refine[44] relies on other LLMs to provide feedback and guide improvement.

Planning

Planning decomposes complex tasks (for example, scientific discovery) into more manageable subtasks and pursues promising paths toward a final solution[45,46]. A variety of techniques address planning from different perspectives. One approach completely decomposes a task into subtasks before solving them sequentially, as seen in least-to-most prompting[47] and ReWOO[48]. Alternatively, some methods decompose and solve subtasks dynamically, addressing each subtask in turn before proceeding to the next, as in ReAct[49] and HuggingGPT[50]. The latest methods, such as AdaPT[51], perform as-needed planning, decomposing a task only when the system encounters failures.

Symbolic planning decomposes tasks and presents action plans in a human-readable format, as is done in LLM + P[52]. It also enables the incorporation of explicit rules and ethical guidelines for high-stakes biomedical research.

Planning is a learnable skill and can be trained using RL to identify the most rewarding actions given the state of a task. AGILE[53] teaches LLMs how to use various tools at their disposal, such as reflection, document retrieval, memory recall and seeking human experts' advice.

Overall, it is often difficult to anticipate the complexity of biomedical research until experimentation is underway; therefore, dynamic planning is necessary for biomedical applications of agentic AI.

Tool use

Biomedical research requires proficiency in specialized bioinformatics tools such as PLINK[54], machine learning (ML) tools such as scikit-learn[55] and the Tree-based Pipeline Optimization Tool[56,57] and biomedical knowledge graphs such as Hetionet[58]. Tool use distinguishes agentic AI from traditional AI models, as agentic AI can determine how and when to use these specialized tools for a given task.

Some methods focus on creating an agent equipped with the appropriate tools for a specific task, much like recruiting a team member with the right skill set[59]. Other techniques aim to build frameworks that support diverse and flexible tool use[60–62]. For example, ToolFormer[60] uses a self-supervised framework that iteratively learns when and which external tools to call before providing the final answer, thereby reducing the need for human intervention in selecting the appropriate tool. The development of such tool-use

frameworks has practical implications for how the biomedical community can augment or fine-tune AI agents to better manage diverse biomedical tools.

Retrieval-augmented generation (RAG[63]) is an important class of tools that can be integrated into an agent to affordably update the static knowledge embedded in LLMs[64]. A RAG module retrieves relevant documents and augments the input prompt with updated knowledge before generating a response[65]. Particularly, the recognition of biomedical entities (for example, gene symbols) is not guaranteed by default. Methods such as Self-BioRAG[66] improve RAG accuracy in retrieving biomedical documents by further training an LLM using clinical and molecular instruction datasets. Tool use enables biomedical applications of agentic AI to run bioinformatics and data analysis tools on behalf of researchers, thereby allowing researchers to focus on high-level tasks.

Memory

Memory in agentic AI generally refers to contextual information or summaries of past events that can be stored and retrieved by the system. Memory helps prevent agentic AI systems from forgetting essential facts or repeating failed experiments. Agentic AI systems, however, do not have an inherent memory mechanism of past events. Consequently, they may lose context and begin providing unrelated responses during extended conversations. A common solution is to append dialog history to user prompts as short-term memory, at the expense of increased token usage and inference latency.

To address the limitations of short-term memory, several types of long-term memory (LTM) have been proposed to support more efficient information recall. For example, some approaches maintain an internal summary of dialogs[67], while others mimic human LTM by selectively forgetting unimportant or outdated events[68,69].

Agentic AI systems can store long-term task-related memory as log entries in external databases and long-term factual knowledge in external knowledge bases. This structured memory enables knowledge retrieval from reliable sources[70] and can facilitate faster responses, instead of slower reasoning[71]. For example, GraphRAG[72] organizes the retrieved documents into a graph database of related text chunks, ensuring that the LLM is not overwhelmed by the volume of retrieved information.

Overall, efficient memory management can reduce token usage and system latency, especially on resource-limited consumer hardware. However, memory alone does not necessarily reduce LLM hallucination or prevent context drift during lengthy conversations. Ensuring the reliability and cost-effectiveness of agent memory remains a critical challenge for the real-world deployment of biomedical agentic AI[73].

Agent–agent, agent–human and agent–tool communication

Efficient communication among agents, humans and computers significantly influences the overall quality of work produced by agentic AI systems, analogous to the role of communication in human teams. With the rise of multi-agent AI systems, several agent-to-agent communication techniques have been developed to enhance communication quality from various perspectives[74]. ChatDev[75] implements a back-and-forth chat chain, allowing agents to request clarification from instructors as needed. MetaGPT[76] deploys a shared message pool to standardize and encourage information exchange among agents. GPTSwarm[77] models AI agents as nodes in a network and communication among agents establishes edges; this design allows the use of graph-based algorithms to optimize information exchange. These approaches resemble communication strategies employed in a biomedical research group, such as clarifying responsibilities and standardized procedures.

To improve interoperability among agentic AI systems, an open protocol called Agent2Agent[78] standardizes secure and task-centric communication among agents or across agentic AI systems, even without shared memory, tools or context.

Human–agent interaction is an extension of human–computer interaction for agentic AI, which aims to respect the rights and values of human researchers throughout the system's decision-making process[79]. For example, a human-centric interactive planning framework has been proposed that invites users to interrupt time-consuming processes or rectify intermediate outputs[80].

Agent-to-computer or agent-to-tool information exchange within an agentic AI system can be optimized using methods such as the Model Context Protocol (MCP[81]). MCP is an open communication framework that standardizes the way agentic AI systems access local or external computational resources. Building on this standard, community-based and open-source initiatives such as BioContextAI[82] employ MCP to provide structured, standardized access to validated scientific sources, thereby fostering a growing ecosystem of tools, knowledge bases and data sources as MCP servers for biomedical agentic AI. The security of MCP for data-sensitive biomedical research, however, remains an open question[81].

To conclude, effective communication protocols ensure reliable interactions among the components of an agentic AI system, which is essential in an inherently collaborative and data-intensive field such as biomedical research.

Current biomedical applications of agentic AI

Agentic AI systems have found applications in various domains such as debugging[83] and full-cycle development[75] in software engineering and disease diagnosis[84,85] and informed clinical decision-making[86] in healthcare. In biomedical studies, researchers engage in a range of iterative tasks that agentic AI systems can facilitate, including literature review, figure comprehension, hypothesis generation, experimental design and data analysis. Designing agentic AI systems for these research tasks resembles assembling a team of individuals with the expertise needed for a job (Fig. 1b). Here we discuss agentic AI systems that automate key steps in the research process.

Literature review

A literature review lays the foundation for scientific discoveries but is often a time-intensive process (Fig. 2a). Several agentic AI systems address this challenge by automating literature retrieval and information extraction. For example, SciLitLLM[87] is a suite of LLMs that aims to acquire an enhanced understanding of scientific literature by a combination of continual pretraining and supervised fine-tuning on high-quality scientific literature. Another approach combines biomedical knowledge graphs and LLMs to ensure accurate keyword recognition and relevant document retrieval[88].

Scientific figures often convey more nuanced information than text. Accordingly, continuous efforts in 'figure-to-text' translation adapt general multimodal LLMs for biomedical figure comprehension[89,90].

Agentic AI systems do not eliminate a researcher's responsibility for a thorough literature review; rather, they function as assistants that automate the time-consuming process. For example, prompt engineering can be used to anchor an AI agent within a specific research context; a RAG–based reviewer agent can support iterative literature discovery by updating its retrieval and summary based on interim findings; and knowledge-graph-augmented AI agents can narrow or broaden searches by identifying publications connected through conceptually similar or distant keywords.

Hypothesis generation

Scientific hypotheses test possible explanations for real-world observations (Fig. 2b). One evaluation presents a multi-agent AI system that generates and continuously refines biomedical hypotheses based on rounds of literature retrieval; these hypotheses are further evaluated for relevance, newness, feasibility and significance[91]. AI Co-Scientist shares a similar architecture but includes other specialized agents, such as an agent that ranks hypotheses[92]. In contrast, iKraph[93]

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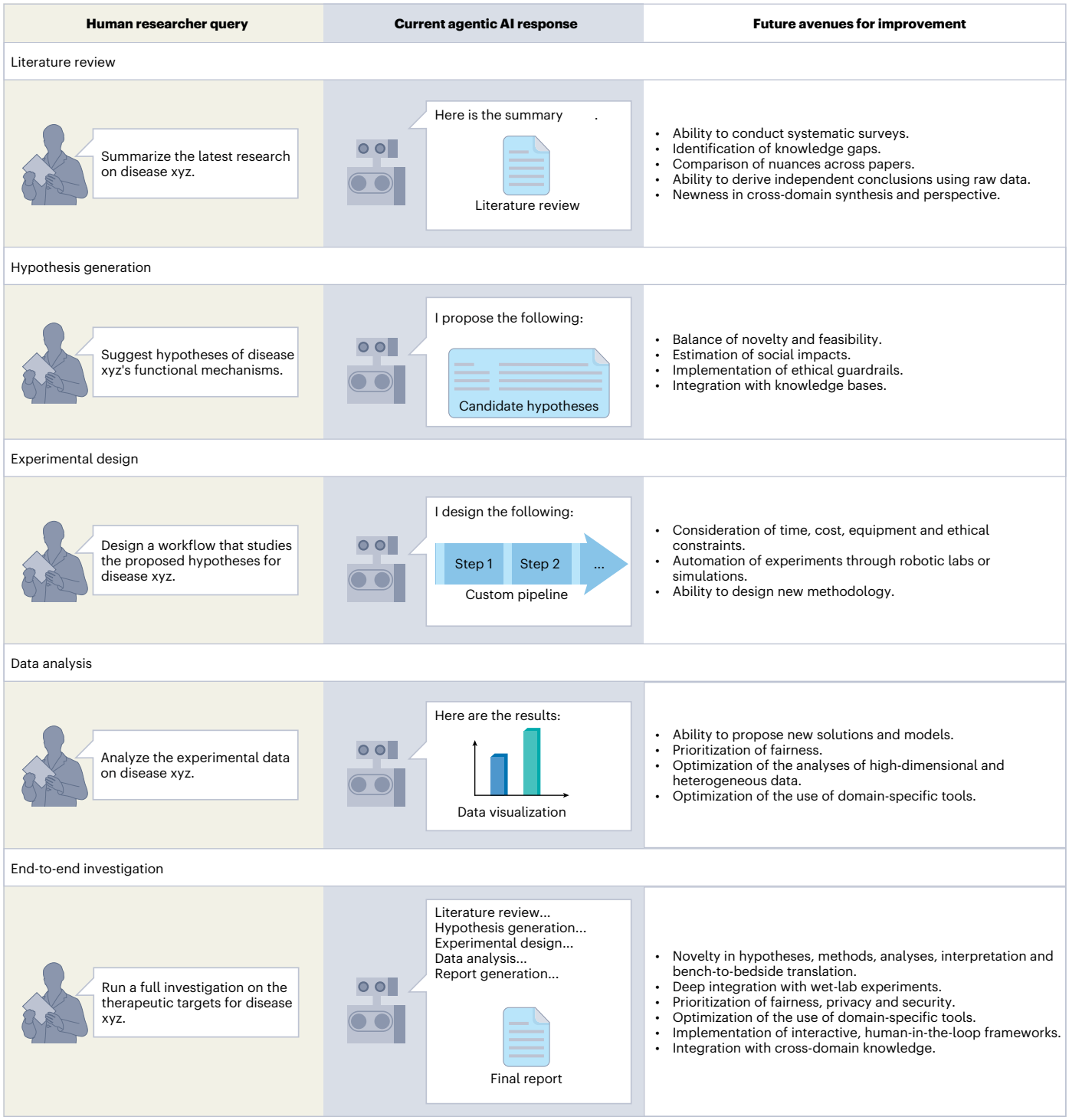


Fig. 2 | Applications of agentic AI systems in various biomedical research tasks. Existing agentic AI systems can perform major biomedical research tasks, rivaling human performance in tasks such as literature review, hypothesis generation, experimental design, data analysis and end-to-end investigation. The capabilities of existing agentic AI systems demonstrate how agentic AI can act as an assistant in biomedical research (second column). We anticipate future agentic AI systems to be improved in various avenues, as listed in the figure (last column). Eventually, agentic AI systems are poised to become not only assistants but allies in biomedical research, advancing further in autonomy and newness. SMK, serine–methionine–lysine; MSK, methionine–serine–lysine; VTV, valine–threonine–valine.

constructs a biomedical knowledge graph from 34 million PubMed abstracts, enabling probabilistic semantic reasoning that automatically generates hypotheses.

Latest proposed agentic AI systems, such as POPPER[94], automate the validation of AI-generated hypotheses by decomposing them into testable subhypotheses. Through a series of falsification experiments, POPPER rejects AI-generated hypotheses if any subhypothesis is proven false.

Experimental design

In addition to planning, experimental design requires a deep understanding of laboratory protocols and specialized analytical tools

(Fig. 2c). BioPlanner[95] represents biological laboratory protocols using textual pseudocode. BioDiscoveryAgent[96] designs gene perturbation experiments by dynamically proposing the next candidate drug targets based on established biological knowledge and the latest experimental results.

Data analysis

Agentic AI systems can expedite biomedical research by conducting end-to-end analysis pipelines, especially for experiments that heavily rely on programming (Fig. 2d). DrugAgent[97] automates programming for drug discovery by implementing a planner that explores potential research ideas, an instructor that translates an idea into programming tasks and a coder that executes and debugs the code. GenieAgent[98] learns preliminary references that match human researchers' intentions with specific tools, thereby improving the accuracy of tool use and automated data analysis for molecular design.

Some agentic AI systems accelerate data analysis by optimizing computational methods for domain-specific tasks. For example, CellForge[99] generates optimized model architectures and executable code for training virtual cell models, given raw data (for example, single-cell multi-omics data) and task descriptions. TusoAI[100] organizes domain knowledge into a knowledge tree that guides the iterative refinement of domain-specific methods, such as denoising methods for single-cell RNA-seq data.

End-to-end research cycle

The success of LLMs in research subtasks has facilitated the development of a variety of agentic AI systems for end-to-end biomedical research (Fig. 2e), including Biomni[101], Sakana's AI Scientist[102,103] and Virtual Lab[59]. Specifically, Virtual Lab[59] is an experimental LLM-based multi-agent system that aims to address interdisciplinary scientific research questions. In Virtual Lab, a human researcher defines the research goal and provides high-level feedback, while a principal investigator agent coordinates a group of scientist agents. Virtual Lab demonstrates its ability to conduct drug development research by successfully identifying 92 new nanobody designs for SARS-CoV-2 variants. These results highlight how an LLM's intrinsic domain knowledge and reasoning capacity can offer exciting opportunities for agentic AI research assistants.

Overall, agentic AI systems are no longer limited to tool-like applications and are now capable of performing essential research tasks as assistants. We anticipate future agentic AI systems to function more as a research collaborator, working closely with human researchers to spark biomedical discoveries (Fig. 2).

Biomedical benchmark suites

Benchmarking assesses the efficacy of agentic AI systems by evaluating their capabilities and limitations in a controlled environment before extending them to real-world applications[104,105]. Most benchmark suites evaluate general utility, such as abstract reasoning[21], alignment with human expectations in daily tasks[106] and the ability to build end-to-end ML projects[107]. However, conventional benchmark suites may fail to capture the nuanced complexity and domain-specific challenges that characterize biomedical research. As such, benchmark suites tailored for biomedical research should assess a system's proficiency in conducting end-to-end scientific workflows, including writing domain-specific code, executing biomedical analysis pipelines, interpreting and summarizing scientific literature, generating testable hypotheses and managing both laboratory and computational resources.

While benchmark suites exist for general and scientific coding tasks[108–111], there is a need for benchmark suites specifically tailored to programming tasks in biomedical research. Code in biomedical domains frequently depends on specialized software packages (for example, PLINK[54]) that are rarely used outside the field and may lack adequate online resources. Consequently, dedicated benchmark datasets are required to evaluate domain-specific code-writing and tool use.

Several benchmark suites evaluate agentic AI systems' critical thinking capabilities that stem from domain-specific knowledge and analytic interpretation of scientific literature. BIG-Bench[112] assesses understanding of scientific concepts across diverse fields, including biology, chemistry and physics. MedQC[113] and MedMCQA[114] focus specifically on medical concepts. AAAR-1.0[115] and PubMedQA[39] evaluate an agentic AI system's comprehension and critical analysis of scientific literature, such as its ability to identify weaknesses and accurately extract conclusions from biomedical papers.

Hypothesis generation is a crucial, yet time-intensive, component of biomedical research. Several initiatives have used labeled data or real-time human reviews to evaluate the creativity and feasibility of AI-generated biomedical hypotheses[90,115,116]. For example, SciFact[116] contains 1.4 k expert-written biomedical claims supported with labeled annotations.

Biomedical agentic AI systems and their associated benchmarking methodologies are still in the early stages of development. Preliminary efforts, such as LABBench[117] for molecular biology and AstaBench[118] for computational biology, demonstrate the potential of standardized end-to-end evaluation. However, biomedical research encompasses a diverse set of subfields, each requiring specialized domain knowledge, distinct research protocols and tailored analytical tools. This diversity underscores the need for more expert-curated, domain-specific benchmarks to assess the effectiveness and reliability of agentic AI systems across biomedical research areas.

Challenges for biomedical applications

The application of agentic AI systems in biomedical settings introduces several unique challenges. Here we discuss five key challenges that warrant further consideration.

Data formatting, dimensionality and heterogeneity

Data constitute the foundational basis for learning in most ML and AI systems, enabling these models to infer, generalize and generate patterns, outputs and behaviors. Consequently, data serve as the primordial building blocks from which complex agentic systems emerge, provided they are effectively captured, structured and integrated within diverse AI and ML frameworks. In biomedical research, however, the primary challenges surrounding data often arise not only from sparsity but from the difficulty of processing, curating and leveraging it efficiently within computational paradigms.

In biomedical research, data are typically decentralized and distributed across multiple institutions, each employing distinct collection and storage formats. This variability and fragmentation complicate the pipeline from data acquisition to model training, requiring researchers to develop effective strategies for harmonizing and processing diverse datasets. To address these challenges, researchers have proposed various data standards. Standardized metadata specifications, such as Croissant[119] govern dataset-level descriptors, including structure, sources, splits, features and ethical considerations. Healthcare-specific common data models and nomenclature systems standardize data relationships and vocabularies, respectively[120]. Together, these data standards help reduce redundant interinstitutional data harmonization and facilitate interoperability. However, the rapid evolution of biomedical science continues to generate new, distinct data types, structures and ontologies. Data standards may fail to capture critical contextual information, such as laboratory measurement error, phenotyping error and natural data drift. Consequently, fine-grained data standardization and harmonization remain major challenges in processing biomedical data.

Biomedical data is also inherently heterogeneous and high-dimensional, spanning modalities such as medical imaging, electronic health records and genomic data. Electronic health records are inherently heterogeneous, containing diverse data types such as clinical

notes, laboratory test results (for example, blood tests and glucose levels) and demographic attributes (for example, age and income). Genomic data illustrate the extensive dimensionality, often comprising millions of single-nucleotide polymorphisms. These challenges are further complicated by growing data volume and by inaccuracies resulting from technical and human error.

Data heterogeneity and high dimensionality pose two primary challenges for researchers—the need to effectively process, engineer and extract relevant features across diverse data modalities and the difficulty of identifying optimal methods to integrate these features into a model. Agentic AI systems offer promising solutions for addressing both challenges. For example, one AI agent can specialize in identifying the most effective computational tools to preprocess and analyze a specific biomedical dataset, while another agent can focus on integrating multiple heterogeneous data sources into a unified representation suitable for subsequent analysis. The ability to create specialized agents within an agentic AI system reduces the manual effort associated with data cleaning and standardization, enabling researchers to focus on higher-level analytical and interpretive tasks rather than the burdens of data preprocessing.

Privacy and security

The integration of agentic AI into biomedical research demands careful consideration of privacy and data security, especially when using sensitive patient data[121]. Agentic AI systems must be designed to meet high data protection and security standards. Failure to uphold these standards can result in severe consequences, including breaches of patient trust, regulatory violations and potentially catastrophic impacts on healthcare systems. For example, in many European countries, the use of personally identifiable information (PII), such as health and demographics data, is regulated by the General Data Protection Regulation (GDPR[122]). Under that law, misuse of PII is considered a high-risk violation for which defaulting organizations can be fined millions of euros. Notably, the GDPR protects European Union citizens' PII even when it is processed outside the European Union. In the United States, the Health Insurance Portability and Accountability Act[123] focuses on how organizations handle protected health information.

Another aspect of privacy relates to individuals' right to control the use of their personal data. Several countries have enacted laws to safeguard this right. For example, under GDPR[122], individuals have the right not to be subject solely to automated decisions that can have legal effects on them. Additionally, although not a binding federal law, the US Office of Science and Technology Policy recently released the Blueprint for an AI Bill of Rights[124], which states that persons should be able to opt out of automated systems, where appropriate and have access to a human alternative. Although these laws exist to protect personal data, they might need to be updated in light of advances in generative AI.

LLMs exhibit well-documented privacy and data security vulnerabilities due to their tendency to memorize and leak sensitive training data[125,126]. Security threats can induce harmful outputs or degrade model performance by manipulating LLM prompts or training data. Privacy attacks exploit the model's internal representations to retrieve confidential training data. Mitigating these vulnerabilities is critical for the secure and privacy-preserving deployment of LLM-based agentic AI systems in biomedical research. One approach to mitigating privacy concerns is to train foundational models on publicly available databases such as the GWAS Catalog[127] and the GenBank genetic sequence repository[128], while ensuring that private databases continue to protect patient-level data. Such an approach not only reduces the risk of privacy breaches but also promotes transparency, reproducibility and broader accessibility in biomedical AI research.

Cost, energy and hardware

Training an LLM from scratch can be a time-consuming and costly undertaking, which often becomes a barrier to entry for laboratories with limited budgets. The availability of open-weight LLMs such as Llama 4 and DeepSeek-V3.2 (ref. 8) can alleviate some of the issues encountered during training. However, the hardware requirements and energy costs remain high during inference. Energy consumption during inference depends on the specific LLM and the response generated corresponding to a given prompt. Smaller LLMs (for example, codellama-7b) use substantially less energy to generate one token than larger ones (for example, codellama-70b)[129,130] and the longer responses are highly correlated with high energy consumption[129]. The energy cost of agentic AI systems not only makes them expensive but also has adverse environmental impacts, including carbon[131] and water[132] footprints. Although hardware architectures can be optimized to use less energy while delivering equivalent performance[133], reducing the carbon and water footprints of these technologies remains an open challenge.

Fairness

Conceptually, algorithmic fairness is defined as the ability of a given model to perform equivalently for different subgroups within the population specified in the dataset[134]. Models that fail to meet this requirement are considered unfair or biased. Unfair AI models exacerbate pressing concerns in healthcare, including the perpetuation of identity-based biases[135] and the aggravation of health disparities in diagnosis, treatment and billing[136]. Fairness has real-world consequences, such as models that recommend unjustified or unnecessary medical interventions[137].

One way to make agentic AI systems fairer is to train and fine-tune them on more representative data. This type of data can come from multiple countries and regions and from people of diverse backgrounds, such as different racial and ethnic groups. This can be challenging as different countries and regions have different privacy protection laws, digital health regulations and ethical frameworks that collectively govern the access, sharing and secondary use of health data for AI and research[138]. Certain strategies, such as federated learning[139,140], may be useful in these contexts. This type of learning involves training a model across multiple clients (for example, hospitals) without centralizing raw data. A central server sends a global model to local clients, which train the model on their private data. Only model updates, rather than the underlying data, are transmitted back to the server and aggregated to produce an improved global model.

Reliability

As agentic AI systems become more sophisticated, failure assessment becomes increasingly complex. One proposed categorization classifies failures into the following three categories: flawed AI architecture (for example, ambiguous agent roles and responsibilities), ineffective or conflicting agent teamwork and poorly defined research questions or termination standards[141]. These failures necessitate improvement in system design, the formulation of scientific questions and human collaboration with agentic AI systems.

Opportunities

Biomedical research is an interdisciplinary field that addresses increasingly complex and computationally intensive health questions. Agentic AI offers potentially transformational opportunities by accelerating repetitive research cycles and sharing the cognitive workload with human researchers.

Knowledge base

The effectiveness of agentic AI systems depends on the extent of domain expertise embedded within each agent. The integration of knowledge graphs enables agents to gain deeper contextual understanding and improves their effectiveness in domain-specific, complex analytical tasks. RAG[63], when combined with existing knowledge graphs[72], provides strategies to enhance agents' decision-making capability.

For example, ESCARGOT[142] combines GraphRAG[72] with the Alzheimer's Knowledge Base[143], reducing hallucination and improving the performance of an agentic AI system on multihop reasoning questions related to Alzheimer's Disease. BioChatter[144] is another community-initiated, open-source agentic AI system that extends BioCypher-constructed knowledge graphs[145] and facilitates the development of custom biomedical research software.

Biomedical research regularly processes various data types (for example, medical imaging and protein structure data). Multimodal LLMs can facilitate the incorporation of multimodal biomedical knowledge bases into agentic AI systems[146].

Biomedical programming

Although LLMs have demonstrated the ability to be effective programmers for general programming problems[75,83], this ability may not easily translate to biomedical research. Future opportunities lie in training specialized AI agents for coding and software development geared toward biomedical research, including domain-specific tools (for example, PLINK[54]) and programming languages (for example, R). Ideally, researchers would not be prevented from exploring specific research ideas due to limited knowledge of specialized tools.

Ethical alignment and social impact

RLHF enhances an LLM's safety and reliability by adhering to explicit constraints, executing correct instructions, reducing hallucinations and employing appropriate language[17]. The ability to fine-tune agentic AI behavior to align with human preferences—commonly referred to as value alignment—and with ethical principles[17,23,147] makes RLHF a desirable tool for high-stakes biomedical research. Additionally, AI agents should leave ample room for human users to retain meaningful autonomy to prevent excessive dependence on these systems[148]. This can include, among other things, evaluating the long-term impacts of using agentic AI systems through randomized control trials.

The use of generative AI technology within agentic AI systems can have some adverse impacts on human users. For example, individual users tend to use less critical thinking (for example, recalling, combining and judging ideas) when using generative AI tools, particularly when they have more confidence in the tools than in their own ability to solve the tasks at hand[149]. Evidence of generative AI's impact on workplace productivity is mixed and inconclusive. One survey reported average productivity gains of 15–30% in real-world field experiments, with the greatest improvements observed in well-defined, structured tasks[150]. As tasks become more complex and require nuanced expert judgment, productivity gains become highly context-dependent. In a randomized controlled trial, for example, experienced software developers spent 19% more time completing tasks, contrary to pretrial expectations of a 20–40% speedup[151]. While generative AI can narrow the experience-based performance gap for entry-level workers, recent research surveys, platform payment data and administrative records indicate a notable decline in demand for entry-level or novice roles in AI-complementary occupations[150].

Interaction with user devices

Biomedical research involves direct interactions with computing devices (for example, laptops) and software applications (for example, web browsers), such as for literature reviews and the use of integrated development environments for data analysis. The latest research aims to grant agentic AI systems visual access to devices and the ability to securely manipulate user applications. For example, the SIMA team demonstrates how these AI agents can read graphical interfaces and translate natural-language instructions into mouse and keyboard actions[152]. Similar principles have incentivized commercial applications such as ChatGPT Operator and the Perplexity Assistant. Device-interacting AI agents have the potential to serve as research assistants that directly engage with user devices and execute end-to-end research workflows, freeing researchers to focus on higher-level tasks such as critical thinking and project supervision.

Creating agentic AI systems for biomedical research

If a user seeks to use an agentic AI system in their research, it may be advantageous to build on an established system, given the diversity of existing scientific agentic AI systems. Alternatively, the development of an agentic AI system from scratch may be guided by three recommended steps.

The first step is to define the overarching goal of the agentic AI system, such as performing end-to-end research automation. One should also envision the subtasks the system will perform—whether in parallel, sequentially or iteratively—to achieve the said goal. Examples of such subtasks include conducting parallel literature reviews on various relevant topics, iterative hypothesis refinement and sequential data analysis. This step clarifies the system's intended scope and provides the foundation for subsequent steps.

The second step concerns identifying the minimal functional requirements necessary to accomplish the specified task. This may include determining the appropriate number of agents, integrating specialized tools, implementing LTM and coordinating multi-agent communication. The relevant system characteristics and design considerations are discussed in the section on 'Agentic AI characteristics'.

The third step is to iteratively improve and benchmark the agentic AI system. This process may involve experimenting with various system characteristics and techniques detailed in 'Agentic AI Characteristics', such as selecting appropriate LLMs or LRMs, applying prompt engineering for value alignment, implementing customized tools or retrieving publicly available tools from biomedical MCP hubs such as BioContextAI[82] or ToolUniverse[153]. More advanced adjustments include clarifying agent roles and duties, incorporating self-reflection and ReAct mechanisms, optimizing multi-agent architectures (for example, sequential, loop, parallel or hierarchical) and evaluating system performance using domain-specific benchmarks, as discussed above.

If the final agentic AI system fails to meet the intended objectives, revisiting earlier design steps may be necessary. For instance, one might simplify the overarching goal, further decompose subtasks or employ more specialized agents. Additionally, throughout this process, it is critical to consider the technical, ethical and methodological challenges discussed, particularly those related to environmental impacts, social consequences, transparency and human oversight in the design of agentic AI.

Future prospects for agentic AI in biomedical research

While agentic AI is in its early developmental stage, preliminary proof-of-concept studies highlight its potential application in a wide range of biomedical research tasks (Fig. 2). In recent years, there has been a swift development of agentic AI systems across various biomedical domains, including functional genomics[96], genetic editing[154], drug molecule design[155], antibody engineering[59], spatial genomics[156] and proteomics[157] (Fig. 3). We believe that agentic AI may potentially have additional applications in biomedical domains that have not yet been explored. For example, agentic AI systems could support large-scale 'omics research by providing comprehensive interpretations of experimental results, recommending follow-up analyses, customizing bioinformatics protocols for high-throughput sequencing data and proposing hypotheses about underlying functional mechanisms, including preliminary investigations using large-scale biobanks. In the coming years, we anticipate rapid and widespread adoption of agentic AI systems across diverse biomedical domains as these tools become increasingly accessible and user-friendly.

We anticipate a progressive transition from specialized single-agent AI systems to general-purpose multi-agent AI systems.

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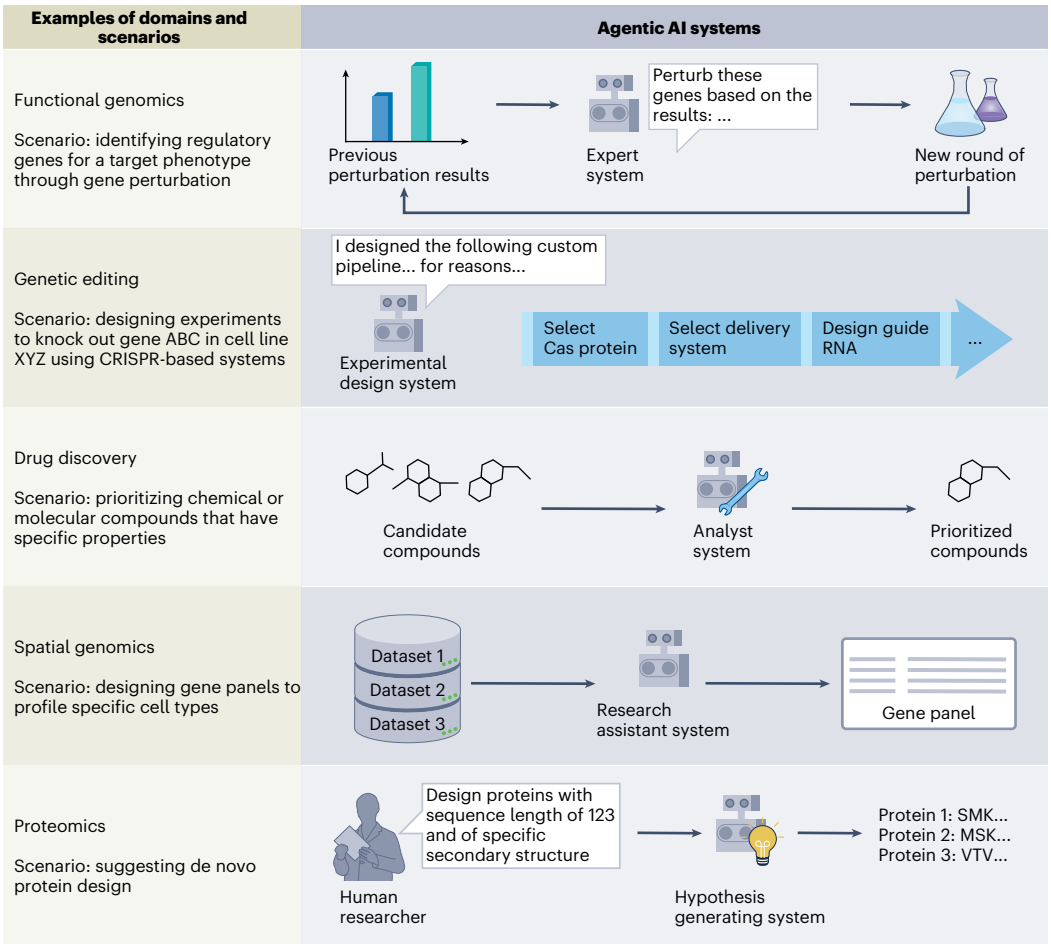


Fig. 3 | Examples of agentic AI systems in various biomedical research domains. Preliminary proof-of-concept studies showcase how agentic AI systems can benefit diverse biomedical domains, including functional genomics (suggesting iterative gene perturbation experiments guided by prior results), genetic editing (composing CRISPR experimental design by selecting Cas protein, delivery system and guide RNAs), drug discovery (prioritizing candidate compounds), spatial genomics (integrating multiple datasets to design targeted gene panels) and proteomics (de novo protein design via hypothesis generation). As agentic AI systems become more user-friendly and accessible, adoption is expected to expand rapidly across additional biomedical use cases.

Single-agent AI systems may be relatively easier to develop, as researchers can focus on optimizing a single framework; such systems can be evaluated rapidly on technically challenging tasks, such as constructing end-to-end ML pipelines. More complex research tasks, such as hypothesis refinement, may require the coordinated collaboration among multiple specialized agents (for example, expert and critic agents; Fig. 1). In fact, multi-agent systems have demonstrated superior performance relative to single-agent systems in some scenarios, such as achieving higher scores on science tests[158] and producing better hypotheses that are judged based on newness[92]. However, establishing effective coordination and communication protocols among multiple agents can be one major challenge for future biomedical agentic AI.

Role assignment is a fundamental component of multi-agent AI systems and generally requires human intervention. One strategy is to design fixed roles tailored to system objectives, based on either human-like traits (for example, individual stock-trading beliefs in TwinMarket[159]) or functional duties (for example, scientific critic and ML specialist in Virtual Lab[59]). Another approach involves the dynamic instantiation of AI agents based on task requirements. For instance, MedAgent[85] generates AI medical specialties based on the disease context provided in the prompt. In the future, we anticipate that more research will focus on techniques for the autonomous assembly and engineering of AI agent teams tailored for specific tasks (for example, Automated Design of Agentic Systems[160] and EvoAgent[161]), thereby reducing the need for human intervention.

Agentic AI systems can have varying levels of autonomy, with one extreme being a fully autonomous AI scientist that conducts experiments and pioneers scientific discoveries without human oversight[162]. While this ability may be appealing, such autonomy might not be appropriate for high-stakes biomedical studies (for example, patient-facing applications). Most agentic AI systems lack training in and understanding of physical constraints and real-world consequences, often proposing creative but infeasible experimental designs or even engaging in problematic and deceptive behaviors[6]. We argue that agentic AI systems can be granted high autonomy for well-defined, rule-based processes, such as customizing standardized data analysis pipelines. However, they should proactively seek human feedback for tasks that require critical thinking (for example, assessing the feasibility of an experimental design) or human judgment (for example, evaluating patient harm or ethical implications). In other words, rather than pursuing full autonomy, we advocate for adaptive autonomy—systems capable of effectively understanding when to consult human experts for ambiguous or risky tasks.

Community-developed agentic AI systems[101,144,145,153,163] are led by academic institutes, non-profit organizations or consortia and are characterized by a focus on transparency, open access, ethical alignment and strict regulatory compliance. These values hold the key to public trust and scientific innovation. However, community-developed systems face limited hardware resources and volatile funding. In contrast, industry-based agentic AI systems[92,102,103]

leverage substantial computational resources and mature engineering pipelines, which allow them to lead in scalability and real-world deployment. Proprietary systems, opaque interpretability and profit-driven bias, however, can constrain industry-based systems. With federated and privacy-preserving agentic frameworks, collaborations with hybrid community-industry partnerships can combine the advantages from both sides, aligning open governance with scalable deployment.

The successful integration of agentic AI into routine research requires a substantial investment from the biomedical community. Adapting agentic AI systems to specific research areas demands the development of carefully designed, domain-specific benchmarks, which can only be provided by domain experts. During long-horizon research workflows, agentic AI systems rely on human researchers for oversight, feedback and necessary intervention to ensure scientific validity. Moreover, agentic AI systems cannot perceive real-world consequences. Human researchers remain irreplaceable for enforcing ethical guardrails and ensuring responsible deployment. Practical considerations, such as computation cost, energy consumption, resource allocation and environmental impacts, ultimately require human oversight. Thus, human researchers will continue to serve indispensable roles in designing, implementing and governing agentic AI systems for future biomedical research.

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Author contributions

J.H.M. conceived the idea for this study and supervised the project. B.L. wrote the first draft of the manuscript. All authors contributed to the writing, revised the manuscript and approved the final version of the manuscript.

Competing interests

The authors declare no competing interests.

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